

EE364B Progress Report

Automotive Model Predictive Control

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1 Introduction

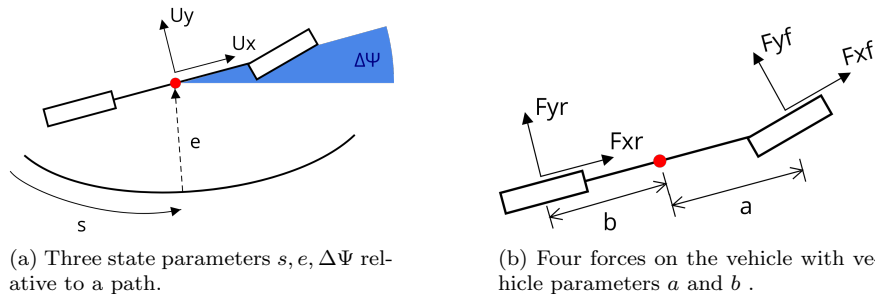
The Dynamic Design Lab at Stanford conducts research on Vehicle Dynamics and Control at the limits of Handling. A former member of the lab, Jonathan Beale, published research in 2013 outlining a control framework that applied Model Predictive Control of vehicle handling at the limits of stability.[1] More recently another member of the lab, John Alsterda, published a research extending this control framework to another framework he referred to as Contingency Model Predictive Control.[2] The basic concept behind this extension was to plan over several trajectories during the optimization step of MPC in order to anticipate errors in random variables in the measurements. This extension was meant to serve as an alternative to Robust Control that is less computationally intensive.

Traditionally a driver has two control inputs that they can use to manipulate a car through space: steering angle δ and drive force Fx . Both formulations of MPC use a linearization of the vehicle model that is non-convex in both the steering angle and drive force. They therefore linearize the vehicle model around a fixed Fx and solve only for the steering input at each step. It seems to me like there is an opportunity for better handling performance if the controller is able to plan using both steering input and drive force – intuitively if you’re trying to corner at high speeds you want to break into a turn and accelerate out of it. Thus my original intent with this project was to research convex relaxations of the vehicle model that might allow for actuation of both drive force and steering angle.

As I’ve developed this project and learned more about Model Predictive Control, I’ve grown pessimistic that I’ll be able to find such a convex relaxations. I’ll explain in more detail later in the progress report. At the end of this report I’ll suggest slight changes to my research direction that I believe may be more feasible.

2 Vehicle Model

We model the vehicle dynamics using the *dynamic bicycle model* pictured in figure ?? . This model assumes that the vehicle has two tires and is constrained to movements along the flat plane of the Earth’s surface.



Given that the vehicle is constrained to a flat plane, it has three degrees of freedom for us to chose. We describe some desired path parameterized by points at $p_i = (s_i, \kappa_i)$ where s is distance along the path and

κ is curvature ($1/R$). We select these degrees of freedom to be s the distance along the path the vehicle has traveled, e the lateral deviation of the vehicle from a fixed path, and $\Delta\Psi$ the difference between vehicle heading and the tangent of the path. We assume that the vehicle can measure it's lateral velocity U_x , it's longitudinal velocity U_y , and it's yaw rate r .¹ We can related these measurements to the evolution of the vehicle state by

$$\dot{s} = U_x \cos \Delta\Psi - U_y \sin \Delta\Psi \approx U_x - U_y \Delta\Psi \quad (1)$$

$$\dot{e} = U_y \cos \Delta\Psi + U_x \sin \Delta\Psi \approx U_y + U_x \Delta\Psi \quad (2)$$

$$\Delta\dot{\Psi} = r - \kappa U_x \quad (3)$$

$$\dot{U}_x = \frac{F_{xf} + F_{xr}}{m} + r U_y \quad (4)$$

$$\dot{U}_y = \frac{F_{yf} + F_{yr}}{m} - r U_x \quad (5)$$

$$\dot{r} = \frac{a F_{yf} - b F_{yr}}{I_z} \quad (6)$$

Previous work published by the DDL uses a linearized tire model called the Fiala Tire model to relate the lateral forces to the steering angle via

$$F_{yf} = -C_{\alpha f} \left(\frac{U_y + ar}{U_x} \delta \right)$$

$$F_{yr} = -C_{\alpha r} \left(\frac{U_y + br}{U_x} \right)$$

where $C_{\alpha f}$ and $C_{\alpha r}$ and the front and rear cornering stiffnesses of the vehicle tires. This is a nonlinear function that relates the angle of the tire to the lateral force it exerts on the vehicle. However, in normal driving operation it can be approximately linearized so we treat this as a linear parameters.² If we hold U_x constant we can write a set of linear dynamical equations for the state $x = (U_y, r)$ as

$$\begin{bmatrix} \dot{U}_y \\ \dot{r} \end{bmatrix} = \begin{bmatrix} \frac{-C_{\alpha f} - C_{\alpha r}}{m U_x} & \frac{b C_{\alpha r} - a C_{\alpha f} - U_x}{m U_x} \\ \frac{b C_{\alpha r} - a C_{\alpha f}}{I_z U_x} & \frac{-a^2 C_{\alpha f} - b^2 C_{\alpha r}}{I_z U_x} \end{bmatrix} \begin{bmatrix} U_y \\ r \end{bmatrix} + \begin{bmatrix} \frac{C_{\alpha f}}{m} \\ \frac{a C_{\alpha f}}{I_z} \end{bmatrix} \delta \quad (7)$$

which is a linear dynamical system in continuous time in the canonical form of $\dot{x} = Ax + Bu$.

3 Controller

The canonical form of Model Predictive Control is

$$\begin{aligned} & \text{minimize} \quad \sum_{t=\tau}^{t+T} l(x(t), u(t)) \\ & \text{s/t} \quad u(t) \in U, x(t) \in X \\ & \quad x(\tau + 1) = A_d x(\tau) + B_d u(\tau) \\ & \quad x(t + T) = 0 \end{aligned}$$

where U is the set of allowable input values, X is the set of allowable state values, l is some cost function we associate with the state and inputs at each step t , and A_d, B_d are the matrices of the discretized linear dynamical system.[4]³ We can discretize the linear system of equations in 7 by

$$A_d = e^{hA} \quad B_d = A^{-1}(e^{hA} - I)$$

¹I've chosen these measured variables to reflect the measurements available to a car with GPS sensors attached like those in use at the DDL.[1]

²For an in-depth treatment of nonlinear cornering stiffness see Jonathan Goh's treatment of it in his PhD thesis concerning automated vehicle drifting [3]

³This form of MPC shows us why we need to restrict our plant model to an affine plant model – the optimization problem uses it to impose an equality constraint which is only solvable for affine equations.

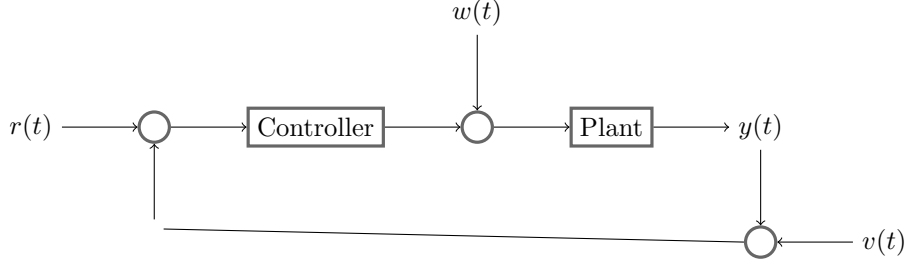


Figure 1: General feedback control structure demonstrating the relationship between the reference input $r(t)$, disturbances $w(t)$, output $y(t)$ and noise $v(t)$.

where h is a constant length of time that separates each point in the MPC plan.[5] At each timestep in MPC the controller solves for the optimal control input over some fixed horizon T and then executes the action u_1^* .

As discussed in section 2 the dynamics of the bicycle model are non-convex (and therefore non-affine) in U_x . A complete controller of the vehicle therefore solves for the speed of the vehicle at each timestep t and sends this as fixed input to the Model Predictive Controller described above to solve for δ . The full control loop is pictured in figure 2.

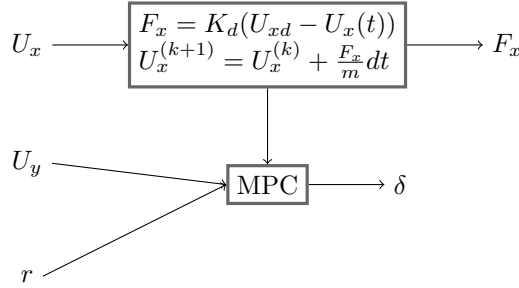


Figure 2: Structure of the control block shown in figure 1. $U_{x\text{desired}}$ is a the speed associated with each point along the path. Nitin Kapania presents a methodology for calculating this desired speed profile in [6].

The cost function can be defined as any convex function of x and u . For simplicity, I am first defining it as a penalty associated with the norms of the optimization variables

$$l(x, u) = \|x - x_{\text{desired}}\|_2^2 + \|u\|_1 \quad (8)$$

Where x_{desired} is a set of points along the path calculated assuming the fixed U_x . Later extensions of this project will likely incorporate penalties on things like the max jerk felt by the car (the first derivative of the acceleration).

4 Simulation Results

As a first milestone of this project I have implemented a modified version of the hybrid Model Predictive - Proportional controller discussed in section 3. For the convex trajectory planning problem I re-used many of the parameters from Craig Beale's paper. [1] Namely, I use a horizon of $N = 15$ and a step length of $dt = 0.01$ [s] and a objective function of the form

$$l(x, u) = \|e - e_{\text{desired}}\|_1$$

The LDS in equation 7 describes the evolution of the lateral velocity, rather than the actual lateral error, so we calculate the lateral error by the approximation

$$e(T) = \sum_t^T U_y(t) dt$$

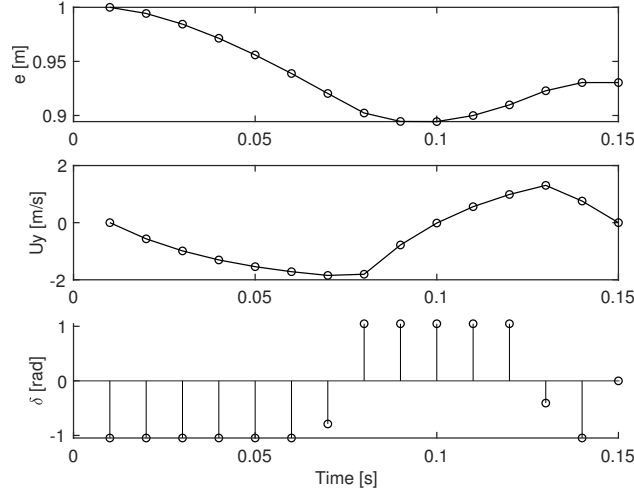


Figure 3: Single trajectory plan for a vehicle at $5 \text{ [m s}^{-1}]$ along a straight path for a initial offset of 1 m.

To confirm the functionality of the basic planning operation of the MPC I first simulated a single planning step with an initial error of 1 [m]. The results of this simulation are plotted in figure 3. Note that the vehicle is only able to decrease the lateral error by about 0.05 [m] over the 0.15 [s] that the MPC algorithm plans over. However, we are only concerned with the first input step before we replan at later times, and this plan confirms that the next input step will not violate any of the constraints imposed.

I've simulated the performance of the MPC over 10 seconds, discretizing over every 100 ms. The results of this simulation are pictured in figure 4. This example is tested on a straight road with an initial lateral error of 1 meter.

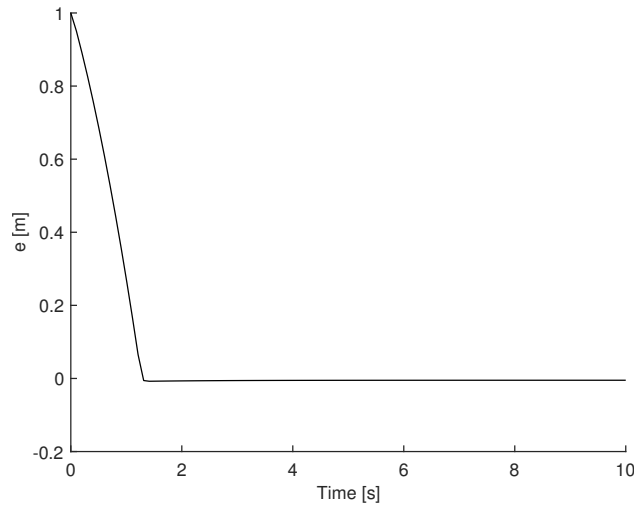


Figure 4: Simulation of the linearized bicycle model using the MPC controller for δ over 10 seconds on a straight road with an initial lateral error of 1 meter.

This solution is INCREDIBLY slow – it took about 30 min to run the simulation for the 10 seconds. One easy opportunity to speed up the computation is to modify the MPC to use a “warm start” or use the previous iteration’s plan as an approximation to the current iteration’s plan.

Nonetheless, this simulation proves that the MPC that I’ve implemented works and that I can now start tweaking the algorithms for performance and a wider scope.

5 Conclusion

I’m way over the page count (sorry) but I’ll close with a few remarks on next steps. So far I’ve implemented an MPC algorithm that mimics the behavior in some of the papers that I’ve cited. [2, 1, 3] This process in itself has been helpful as it has forced me to think through at a granular level exactly what simplifications to the vehicle dynamics we have to make in order to apply a convex optimization framework to controlling the system. The most significant realization that I’ve had is learning that I need an affine approximation to the vehicle dynamics to use in the MPC controller (restricted by an equality constraint).

Next steps involve speeding up the computation time of the controller via warm-start. Further, I also need to extend path tracking to an estimated path calculated assuming a constant U_x . Then, I’d like to extend the work of Alsterda and try to incorporate compensation for disturbing lateral forces (like road grade or something similar). This seems more achievable than coming up with some new representation for the lateral speed that preserves the affine bicycle model required for the MPC equality constraints.

References

- [1] C. B. C. Gerdes, “Model predictive control for vehicle stabilization at the limits of handling,” *IEEE Transactions on Control Systems Technology*, 2013.
- [2] C. G. John P. Alsterda, Matthew Brown, “Contingency model predictive control for automated vehicles,” *American Control Conference*, July 2019.
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- [6] N. K. J. S. C. Gerdes, “A sequential two-step algorithm for fast generation of vehicle racing trajectories,” *Journal of Dynamic Systems, Measurement, and Control*, 2016.